

Project Definition Registration Document

Master of Computer Applications (MCA), Semester III

Submitted By: Abhishek Kamble, Rushikesh Chintanpalli

1. Project Title

- ✓ **Clothy:** Mobile Virtual Try-On (VTON) System with Real-Time Inventory Validation

2. Project Features & Brief Scope

A. Core Features:

- ❖ Staff-assisted Virtual Try-On (VTON) interface accessible via standard mobile browsers.
- ❖ Zero-hardware architecture eliminating the need for expensive in-store smart mirrors.
- ❖ Real-time inventory synchronization preventing try-ons for out-of-stock garments.
- ❖ Generative AI-powered high-resolution rendering of clothing on customer photos.
- ❖ Cloud-optimized image delivery network for fast mobile rendering.

B. Brief Scope:

- ❖ **In-Scope:** Processing men's topwear (shirts, t-shirts) and bottomwear (trousers); real-time database validation; cloud-based AI image generation.
- ❖ **Out-of-Scope:** Public-facing e-commerce payment gateways; complex draping items (womenswear, dresses); live video processing.

3. Data & Resources Availability

- A. **Dataset Sourcing:** High-quality flat-lay and ghost mannequin garment images procured via custom web scraping pipelines (BeautifulSoup/Playwright).
- B. **AI Models:** Utilizing pre-trained, open-source diffusion models (e.g., IDM-VTON, CatVTON) hosted on cloud architecture.
- C. **Documentation:** Technical whitepapers and API documentation for Cloudinary, MongoDB, React, and FastAPI are openly accessible.
- D. **Development Tools:** Visual Studio Code, Postman for API testing, and Git/GitHub for version control.

4. Project Implementation Feasibility

A. Technical Feasibility:

- ❖ High feasibility utilizing a distributed microservice architecture.
- ❖ Frontend built with React.js ensures mobile responsiveness.
- ❖ Backend utilizes standard Node.js/Express.js for inventory logic and Python/FastAPI for AI model execution.

B. Operational Feasibility:

- ❖ BYOD (Bring Your Own Device) approach ensures immediate adoption by retail staff without training or hardware installation.

C. Time Feasibility:

- ❖ Use of pre-trained AI models rather than training from scratch guarantees project completion within the 3rd Semester timeline.

5. Costing for Hardware, Software, Subscriptions, and Cloud

- A. **Hardware:** ₹0 (Utilizes existing store owner smartphones and standard laptops for development).
- B. **Software (IDE/Libraries):** ₹0 (Open-source tools: VS Code, React, Node.js, Python).
- C. **Database (MongoDB Atlas):** ₹0 (Free tier provides sufficient storage for MVP text data and URLs).
- D. **Image Storage (Cloudinary API):** ₹0 (Free tier provides 25 monthly credits, sufficient for development and testing).
- E. **Cloud Hosting (Backend/AI Server):**
 - ❖ Development Phase: ₹0 (Localhost testing).
 - ❖ Deployment Phase (Optional): Approx. ₹2,000 - ₹4,000/month (AWS EC2 / Google Cloud GPU instances if required for live public demonstration).
- F. **Total Estimated MVP Cost:** ₹0 - ₹4,000.